

NOLINSK, Russian Federation

WMO: 273930

Lat: 57.550N

Lon: 49.950E

Elev: 440

StdP: 14.46

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-15.1	-26.8	1.3	-20.9	-20.1	1.8	-14.7	20.9	21.9	19.3	21.9	4.1	0	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	85.0	68.5	81.5	66.6	78.4	65.0	70.2	81.4	68.6	79.0	66.7	75.9	6.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	99.0	75.2	64.7	93.0	73.3	63.0	87.5	71.9	34.3	81.6	33.0	79.8	31.5	75.9	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.9	14.4	-26.8	89.5	6.9	3.6	-31.8	92.1	-35.8	94.2	-39.7	96.2	-44.8	98.8		
(5)			-27.0	73.0	6.9	2.3	-32.0	74.6	-36.0	75.9	-39.9	77.2	-44.9	78.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.1	10.3	11.4	23.5	39.2	53.2	61.5	66.1	61.5	51.1	38.8	24.6	14.2	
	DBStd	22.16	13.39	13.07	8.44	9.02	9.35	7.98	6.81	7.67	7.68	7.72	10.87	13.46	
	HDD50	5870	1232	1081	820	343	77	9	0	3	77	357	763	1108	
	HDD65	10045	1697	1501	1285	775	379	159	69	162	419	813	1213	1573	
	CDD50	1529	0	0	0	19	176	353	500	361	111	9	0	0	
	CDD65	228	0	0	0	0	13	53	105	54	3	0	0	0	
	CDH74	1893	0	0	0	1	138	434	859	439	22	0	0	0	
	CDH80	472	0	0	0	0	21	84	227	138	2	0	0	0	
Wind	WSAvg	6.5	7.0	7.0	7.3	6.5	6.6	5.9	5.0	5.5	5.8	7.0	7.3	7.3	
Precipitation	PrecAvg	24.9	1.9	1.3	1.5	1.3	1.9	2.6	3.0	2.5	2.2	2.8	2.2	1.9	
	PrecMax	30.2	3.3	3.1	3.1	2.9	4.0	5.8	5.5	6.7	5.0	7.7	4.5	3.0	
	PrecMin	19.1	0.6	0.2	0.4	0.3	0.4	0.7	0.4	0.5	0.4	0.7	0.5	0.6	
	PrecStd	3.0	0.7	0.7	0.6	0.7	0.9	1.1	1.5	1.4	1.1	1.3	0.9	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	35.1	44.8	71.6	83.2	86.7	90.4	90.1	77.0	63.8	47.4	38.9	
		MCWB	32.6	33.0	38.7	55.0	64.4	68.9	68.8	68.9	62.0	52.7	45.9	38.5	
	2%	DB	32.8	33.3	39.9	64.6	77.9	82.0	85.3	83.7	71.8	56.8	42.4	35.0	
		MCWB	31.6	32.1	34.8	50.9	60.4	66.6	69.0	68.6	59.8	50.0	41.0	33.9	
	5%	DB	31.0	32.1	36.4	58.5	73.5	79.0	82.2	78.4	67.6	52.7	38.8	33.1	
		MCWB	30.1	31.4	33.0	48.0	58.1	65.1	67.5	65.7	57.9	48.2	37.3	32.1	
	10%	DB	27.9	29.5	34.4	53.3	69.1	75.6	79.0	74.3	63.5	49.8	35.9	31.3	
		MCWB	26.9	28.3	32.0	44.6	55.6	63.6	66.2	63.6	56.1	46.9	34.7	30.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	33.8	39.1	56.1	66.4	71.8	72.7	71.1	64.6	54.8	46.1	38.6
			MCDB	34.3	34.5	44.4	68.3	80.2	82.6	82.8	83.7	72.9	59.9	47.1	38.8
2%		WB	31.9	32.6	35.6	52.3	62.4	68.9	70.6	69.2	61.3	51.4	41.3	34.2	
		MCDB	32.7	33.3	38.4	61.9	73.9	79.1	81.3	81.2	69.0	55.4	42.1	34.8	
5%		WB	30.0	31.4	33.9	48.7	59.7	66.8	69.1	66.8	58.8	49.0	37.8	32.3	
		MCDB	31.0	32.2	35.6	57.8	70.2	76.6	79.6	76.4	65.7	52.0	38.6	33.1	
10%		WB	27.0	28.3	32.5	45.2	57.1	64.6	67.5	64.5	56.6	46.9	34.8	30.4	
		MCDB	27.9	29.3	34.3	52.5	66.7	73.3	76.8	72.7	63.1	49.6	35.7	31.3	
Mean Daily Temperature Range	5% DB	MDBR	10.2	11.8	12.2	14.9	18.1	16.7	16.5	15.5	13.8	9.3	7.4	9.0	
		MCDBR	8.5	9.2	12.6	23.0	23.9	20.6	21.6	21.4	20.6	15.1	8.0	6.3	
		MCWBR	8.4	8.5	9.4	12.3	10.9	9.6	9.2	8.7	10.8	9.8	7.9	6.6	
	5% WB	MCDBR	8.1	8.9	9.9	21.7	21.3	18.8	19.3	19.2	18.7	13.5	7.9	6.3	
		MCWBR	8.1	8.7	8.0	12.3	11.2	9.8	9.3	8.9	10.8	9.8	7.9	6.6	
Clear-Sky Solar Irradiance	taub	0.284	0.312	0.360	0.395	0.379	0.374	0.399	0.411	0.365	0.344	0.303	0.275		
	taud	2.037	2.048	1.984	2.099	2.344	2.397	2.337	2.283	2.387	2.316	2.155	2.010		
	Ebn at Noon	170	221	242	256	272	274	263	250	246	214	164	134		
	Edh at Noon	19	29	41	43	36	35	36	36	28	23	16	14		
All-Sky Solar Radiation	RadAvg	172	449	923	1339	1693	1798	1780	1350	813	351	141	96		
	RadStd	21	78	95	135	152	135	168	134	97	58	23	23		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page